

CHAPTER 4

Establishing hyperbolicity— the magic of cones



Every result obtained in chapter 2 can be extended to a much larger class of systems. The basic tool in chapter 2 has been the very explicit hyperbolic structure of the map. Non linear maps cannot be seen to be hyperbolic by naively analyzing their derivative. Hence the necessity of a more sophisticated tool to establish hyperbolicity.

Our next task is to develop a general approach which allows to establish when the Lyapunov exponents are different from zero almost everywhere.

Although the results presented below look quite general, the reader should be cautioned that their concrete application has been possible only in fairly simple situations, when the distributions are roughly constant on the phase space. The really general case, when the distributions are only measurable and therefore drastically dependent from the geometry of the orbits, is wide open.

4.1 The two dimensional case

We start by dealing with the area preserving two dimensional case in order to explain the basic idea.

Theorem 4.1.1 (Wojtkowski [92]) *Let (X, μ, T) be a dynamical system where X is a compact two dimensional Riemannian manifold, μ is the Riemannian volume,¹ T a diffeomorphism of X and*

$$\int_X \log \|DT\| d\mu < \infty.$$

¹In fact, it is the symplectic not the Riemannian structure that matters; yet, in two dimension, there is not much difference. On the contrary, in section 4.2 this difference will be made explicit.

If there exists a measurable family of convex two sided cones $\mathcal{C}(x) \subset \mathcal{T}_x X$ such that, for almost all $x \in X$, there exists $n \in \mathbb{N}$ with the property²

$$D_x T^n \mathcal{C}(x) \subset \text{int}(\mathcal{C}(T^n x)) \cup \{0\},$$

then the Lyapunov exponents are different from zero almost surely.

A system with a family of cones satisfying the hypotheses of the Theorem 4.1.1 is called *eventually strictly monotone*.³

PROOF. Let $x \in X$ and $n \in \mathbb{N}$ such that

$$D_x T^n \mathcal{C}(x) \subset \text{int}(\mathcal{C}(T^n x)) \cup \{0\}.$$

The first thing to notice is that it is possible to make an orientation preserving change of coordinates (i.e., a change of coordinates via a matrix with positive determinant) both in $\mathcal{T}_x X$ and in $\mathcal{T}_{T^n x} X$ such that, in the new coordinates, $\mathcal{C}(x)$ and $\mathcal{C}(T^n x)$ become the standard cone $\mathcal{C}_+ = \{v \in \mathbb{R}^2 \mid v_1 v_2 \geq 0\}$ and the Riemannian volume is the standard ones (see Problem 4.2). Viewed in this coordinates $D_x T^n$ becomes a two by two matrix with determinant equal to one, that maps $\mathcal{C}_+$ strictly into itself. Note that, since the cone family is measurable, the change of coordinates depends measurably by x .

To continue it is necessary to study a bit the general properties of the matrices enjoying the above mentioned properties. Notice that if we define a quadratic form $\mathcal{Q} : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\mathcal{Q}(v) = v_1 v_2$, then $\mathcal{C}_+ = \{v \in \mathbb{R}^2 \mid \mathcal{Q}(v) \geq 0\}$, so our task is to study the two by two matrices L with $\det(L) = 1$ and such that $\mathcal{Q}(v) \geq 0$, $v \neq 0$, implies $\mathcal{Q}(Lv) > 0$.⁴

Algebraic considerations

Let $v = (1, u)$ with $u \in \mathbb{R}^+$, which implies $v \in \mathcal{C}_+$, and

$$L = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

²A “(two sided) cone,” in a linear space, is a set such that if ξ belongs to the set then $\lambda \xi$ belongs to the set for each $\lambda \in \mathbb{R}$. In addition, we require that the set is closed, has open interior and its complement is non void. Actually, this last conditions could be relaxed for x in a zero measure set without changing the following proof (see footnote 8 at page 111). By “two sided convex cone” we mean that each half cone is a convex set. Notice that, in $\mathbb{R}^2$, such a cone is defined uniquely by the two edges. By measurable we mean that the functions from X to the unit vectors, in the direction of the edges, are measurable.

³In fact, in the case of billiards a similar condition is called *sufficiency* but I find the above terminology more appropriate.

⁴We will call such matrices *Strictly monotone*. Note that if A, B are strictly monotone so are AB and BA .

with $\det(L) = 1$ (note that this is equivalent to L being symplectic, see section 4.2). Then, for each $u \geq 0$, we must have

$$0 < \mathcal{Q}(Lv) = ac + (ad + bc)u + bdu^2, \quad (4.1.1)$$

Setting $u = 0$, it must be $ac > 0$. On the other hand, since $(0, 1) \in \mathcal{C}_+$ and $L(0, 1) = (b, d)$, it must be $bd > 0$. Finally, if we compute the quadratic polynomial in its minimum $u_0 = -\frac{ad+bc}{2bd}$ we get, calling $v_0 = (1, u_0)$,

$$\mathcal{Q}(Lv_0) = -\frac{1}{4bd} < 0.$$

The above relation is possible only if $v_0 \notin \mathcal{C}_+$, which implies $u_0 < 0$ or $ad + bc > 0$, that is $ad > \frac{1}{2}$. Collecting the above results it follows that all the elements of the matrix L must be different from zero and, in addition, they must have the same sign. Since $\mathcal{Q}(v) = \mathcal{Q}(-v)$, without loss of generality we can assume them to be all positive.⁵

The next step is to define some measure of expansion for a strictly monotone matrix. A natural quantity to consider is:

$$\sigma(L) = \inf_{v \in \text{int}(\mathcal{C}_+)} \sqrt{\frac{\mathcal{Q}(Lv)}{\mathcal{Q}(v)}}.$$

Choosing again $v = (1, u)$, it follows

$$\frac{\mathcal{Q}(Lv)}{\mathcal{Q}(v)} = \frac{(a + bu)(c + du)}{u} := f(u),$$

a simple computation shows that f attains its minimum value, for $u \geq 0$, at $u = \sqrt{\frac{ac}{bd}}$. Accordingly,

$$\sigma(L) = \sqrt{bc} + \sqrt{ad} = \sqrt{bc} + \sqrt{1 + bc} > 1. \quad (4.1.2)$$

Moreover, given two monotone matrices L_1, L_2 , we have

$$\sigma(L_1 L_2) = \inf_{v \in \text{int}(\mathcal{C}_+)} \sqrt{\frac{\mathcal{Q}(L_1 L_2 v)}{\mathcal{Q}(L_2 v)}} \sqrt{\frac{\mathcal{Q}(L_2 v)}{\mathcal{Q}(v)}} \geq \sigma(L_1) \sigma(L_2). \quad (4.1.3)$$

An interesting fact, that follows immediately from (4.1.2), is that $\sigma(L) > 1$ if and only if L is strictly monotone.

⁵Just consider the change of variable $x = y, y = x$.

Measure theoretical considerations

The point of measuring the expansion via the $\mathcal{Q}$ -form is due to the following Lemma.⁶

Lemma 4.1.2 *If the Dynamical System (X, T, μ) , X a two dimensional Riemannian manifold, T differentiable a.e. and μ the Riemannian volume, is eventually strictly monotone, then*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \ln \sigma(D_x T^n) > 0 \quad \mu\text{-a.e.}$$

PROOF. Let $\nu : X \rightarrow \mathbb{R}^+ \cup \{\infty\}$ be defined by

$$\nu(x) := \liminf_{n \rightarrow \infty} \frac{1}{n} \ln \sigma(D_x T^n).$$

Then

$$\begin{aligned} \nu(T^{-1}x) &= \liminf_{n \rightarrow \infty} \frac{1}{n} \ln \sigma(D_{T^{-1}x} T^n) \\ &\geq \liminf_{n \rightarrow \infty} \frac{1}{n} \{\ln \sigma(D_x T^{n-1}) + \ln \sigma(D_{T^{-1}x} T)\} \\ &\geq \liminf_{n \rightarrow \infty} \frac{1}{n} \ln \sigma(D_x T^n) = \nu(x), \end{aligned} \tag{4.1.4}$$

where we have used (4.1.3) and the fact that $\sigma(D_\xi T) \geq 1$ by monotonicity.

Let $A_0 = \{x \in X \mid \nu(x) = 0\}$, to prove the Lemma it suffices to show that $\mu(A_0) = 0$. To this end note that, since (4.1.4) implies $\nu(Tx) \leq \nu(x)$, ν and hence A_0 must be invariant.⁷

Consequently, suppose that $\mu(A_0) > 0$, then, for each $m \in \mathbb{N}$,

$$0 = \int_{A_0} \nu(x) \mu(dx) \geq \int_{A_0} \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{[\frac{n}{m}]-1} \ln \sigma(D_{T^{im}x} T^m),$$

where we have used (4.1.3) again. At this point we would like to use BET, yet we face a technical problem: we do not know if $\ln \sigma(D_x T^m)$ is integrable. Nevertheless, we are not interested in large values of $\ln \sigma(D_x T^m)$. It is then natural to define

$$\varphi_m(x) = \min\{\ln \sigma(D_x T^m), 1\}.$$

⁶It is interesting to remark that the next Lemma, together with the results of section 4.3, imply the existence of the stable and unstable distribution (see Examples 4.3.1-Cones and $\mathcal{Q}$ -forms). Thus, since X is two dimensional, the existence a.e. of the L.E. follows as in Problem 3.7 without invoking Oseledec Theorem. The use of Oseledec Theorem is instead necessary in higher dimensions.

⁷Indeed, $\mu(\nu - \nu \circ T) = 0$ by the invariance of μ , which implies $\nu \circ T = \nu$ μ -almost everywhere.

Now $\varphi_m \in L^\infty(X, \mu)$, thus the ergodic average $\varphi_m^+ \in L^\infty(X, \mu)$; hence,

$$\begin{aligned} 0 &\geq \int_{A_0} \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{\lfloor \frac{n}{m} \rfloor - 1} \varphi_m(T^{im}x) = \frac{1}{m} \int_{A_0} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \varphi_m(T^{im}x) \\ &= \frac{1}{m} \int_{A_0} \varphi_m^+(x) = \frac{1}{m} \int_{A_0} \varphi_m. \end{aligned} \quad (4.1.5)$$

That is $\varphi_m = 0$ a.e. in A_0 . But this is a contradiction since, calling $B_m = \{x \in X \mid \sigma(D_x T^m) > 1\}$, the definition of eventually strictly monotone is equivalent to $\mu(\cup_{m \in \mathbb{N}} B_m) = \mu(X)$. Therefore there must exist an $m \in \mathbb{N}$ such that $\mu(A_0 \cap B_m) > 0$, which implies

$$\int_{A_0} \varphi_m \geq \int_{A_0 \cap B_m} \varphi_m > 0,$$

whereby contradicting (4.1.5). $\square$

The relevance of what we have seen up to now for the estimation of the Lyapunov exponents depends on the trivial inequality

$$\|v\|^2 \geq 2\mathcal{Q}(v). \quad (4.1.6)$$

The only real problem left is that, due to our change of variable to put the cones into their standard form, the Euclidean norm $\|\cdot\|$ in the new variables no longer correspond to the original norm in X (let us call such original norm, at the point x , $\|\cdot\|_x$). Nevertheless, the two norms must be equivalent by construction,⁸ hence there must exist an everywhere strictly positive measurable function $a(x) \leq 1$ such that, for each $v \in \mathcal{T}_x X$

$$a(x)^{-1} \|v\| \geq \|v\|_x \geq a(x) \|v\|.$$

Let us introduce the set $A(\varepsilon) = \{x \in X \mid a(x) > \varepsilon\}$, clearly $\cup_{\varepsilon > 0} A(\varepsilon)$ has full measure. Now Poincaré theorem implies, if $\mu(A(\varepsilon)) \neq 0$, that almost all points in $A(\varepsilon)$ return to $A(\varepsilon)$ infinitely often. Let $x \in A(\varepsilon)$ be one of such points, then there exists a sequence n_k such that $T^{n_k}x \in A(\varepsilon)$ for each $k \in \mathbb{N}$.

Accordingly, for each $x \in X$ for which the L.E. exists,⁹ $v \in \text{int}(\mathcal{C}(x))$, $\|v\|_x = 1$, and $m \in \mathbb{N}$ holds

⁸If we admit that $\mathcal{C}(x)$ can have an empty interior on a set of zero measure in X —see footnote 2 at page 108—then the two norms would be equivalent only almost everywhere. Nevertheless, this does not change the proof: call X_1 the incriminated set, then $X_2 := \cup_{n \in \mathbb{Z}} T^n X_1$ has also zero measure and it is an invariant set. We can then discard such a set and work on its complement without any other change in the following.

⁹This is a full measure set by Oseledets Theorem [69].

$$\begin{aligned}
\lambda(x, v) &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \|D_x T^n v\|_{T^n x} \geq \liminf_{k \rightarrow \infty} \frac{1}{n_k} \log \|D_x T^{n_k} v\|_{T^{n_k} x} \\
&\geq \liminf_{k \rightarrow \infty} \frac{1}{n_k} \log \|D_x T^{n_k} v\| + \liminf_{k \rightarrow \infty} \frac{1}{n_k} \log a(T^{n_k} x) \\
&\geq \liminf_{n \rightarrow \infty} \frac{1}{n} \log \|D_x T^n v\| - \lim_{k \rightarrow \infty} \frac{1}{n_k} \log \varepsilon^{-1}
\end{aligned}$$

and, since $\mathcal{Q}(v) \neq 0$ and by (4.1.6),

$$\begin{aligned}
\lambda(x, v) &\geq \liminf_{n \rightarrow \infty} \frac{1}{n} \log \sqrt{\frac{\mathcal{Q}(D_x T^n v)}{\mathcal{Q}(v)}} \\
&\geq \liminf_{n \rightarrow \infty} \frac{1}{n} \log \sigma(D_x T^n) > 0,
\end{aligned}$$

due to Lemma 4.1.2.

We have then seen that, for almost every $x \in X$ and $v \in \text{Int}\mathcal{C}(x)$, $\lambda(x, v) \neq 0$. Next, note that the Lyapunov exponents of T are given by minus the Lyapunov exponents of T^{-1} (see Problem 3.5 and vicinity). Moreover T^{-1} is strictly monotone with respect to $\mathcal{C}(x)^c$. Thus, by Oseledets Theorem 3.2.2 (or see section 4.4), almost every point must have a vector $v_-(x)$ such that $\lambda(x, v_-(x)) < 0$.¹⁰ Obviously, given any vector $v \in \mathcal{TX}$ it follows $\lambda(x, \alpha v + \beta v_-(x)) = \lambda(x, v)$, provided $\alpha \neq 0$, and this concludes the story. $\square$

Remark 4.1.3 *The measurability assumption is a very weak hypothesis but cannot be eliminated. Indeed, if one constructs a cone family along the trajectories it can easily be made strictly monotone. Hence, if the system has zero Lyapunov exponents such a cone family cannot be measurable (see Problem 4.4).*

The above theorem provides us with a very powerful instrument to establish hyperbolicity for a given dynamical system.

To see how it works let us consider some simple examples.

4.1.1 Examples

4.1.1.a Linear automorphisms of the Torus

Consider the matrix

$$L = \begin{pmatrix} 1 & a \\ a & 1 + a^2 \end{pmatrix}$$

¹⁰See Problem 4.5 for an alternative argument.

with $a \in \mathbb{N}$ and the standard cone $\mathcal{C}_+ = \{(v, v) \in \mathbb{R}^2 \mid uv \geq 0\}$. Then

$$L \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u + av \\ au + (1 + a^2)v \end{pmatrix}$$

shows that $\mathcal{C}_+$ is strictly monotone for L . Of course, this is a rather silly example since it is completely obvious that the map is hyperbolic, the next example is a little less trivial.

4.1.1.b *Perturbations of linear automorphisms of the Torus*

Consider a diffeomorphism $\Phi : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ such that

$$\|(D\Phi - \mathbf{1})L\| < 1; \quad (4.1.7)$$

where L is defined as in the previous example, then the map T defined by $Tx := \Phi(Lx)$ is hyperbolic. To see this write

$$DT \begin{pmatrix} u \\ v \end{pmatrix} = L \begin{pmatrix} u \\ v \end{pmatrix} + (D\Phi - \mathbf{1})L \begin{pmatrix} u \\ v \end{pmatrix} := \begin{pmatrix} u + av \\ au + (1 + a^2)v \end{pmatrix} + \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

but $\max\{|\alpha|, |\beta|\} \leq \|(D\Phi - \mathbf{1})L\| \|(u, v)\| \leq u + v$. Thus $\mathcal{C}_+$ is strictly monotone for DT .

It is interesting to notice that, already for this simple example, it would be not immediately clear how to establish hyperbolicity without using a cone language. In addition, remark that the full strength of Wojtkowski theorem it is not used here—since the cone family is strictly monotone.

4.1.1.c *Levowich map*

Let us consider the map $T : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ defined by¹¹

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x - \sin x + y \\ x - \sin x + y \end{pmatrix},$$

It is immediate to verify that

$$D_{(x,y)}T = \begin{pmatrix} 2 - \cos x & 1 \\ 1 - \cos x & 1 \end{pmatrix}$$

Since $\det(DT) = 1$, it follows that $(\mathbb{T}^2, T, m)$ is a Dynamical Systems. In addition, $D_x T \mathcal{C}_+ \subset \mathcal{C}_+$ strictly, apart from the zero measure set $\{x = 0\}$, so Theorem 4.1.1 applies.

¹¹Note that here our torus has the periodicity of 2π instead than one as in the previous examples, this is just to have simpler formulae; the reader can easily reformulate the problem on the torus $\mathbb{R}^2 \bmod 1$.

4.2 Higher dimension—an overview

The difficulties in extending to higher dimensions the previous results stem mainly from the large variety of possible cone shapes in higher dimension. It is far from obvious how to relate monotone properties of a given cone to the behavior of the Lyapunov exponents. One possible way is to generalize the approach based on quadratic forms. We will comment on this possibility in section ???. Yet, in the special case of *Symplectic Systems*, it is possible to develop a very rich theory which is astonishingly similar to the two dimensional one. Here we give a brief insight into this theory, but see [63], [64] and [92] for a much more detailed account.

Definition 4.2.1 *By symplectic Systems we mean a Dynamical System (X, T, μ) where X is a symplectic manifold,¹² μ is the symplectic volume¹³ and T is a symplectic map.¹⁴*

Clearly one can also define a Symplectic Systems in continuous time, we have already seen the typical example: Hamiltonian systems (see Examples 1.1.1).

For the convenience of the reader we will present here some of the material from [94] and [63].

Let $\mathcal{W}$ be a linear symplectic space of dimension $2d$ with the symplectic form ω . For instance we call $\mathcal{W} = \mathbb{R}^d \times \mathbb{R}^d$ the standard linear symplectic space if

$$\omega(w_1, w_2) = \langle \xi^1, \eta^2 \rangle - \langle \xi^2, \eta^1 \rangle, \quad (4.2.1)$$

where $w_i = (\xi^i, \eta^i)$, $i = 1, 2$, and $\langle \xi, \eta \rangle = \xi_1 \eta_1 + \cdots + \xi_d \eta_d$.

The symplectic group $Sp(d, \mathbb{R})$ is the group of linear maps of $\mathcal{W}$ ($2d \times 2d$ matrices if $\mathcal{W} = \mathbb{R}^d \times \mathbb{R}^d$) preserving the symplectic form i.e., $L \in Sp(d, \mathbb{R})$ if

$$\omega(Lw_1, Lw_2) = \omega(w_1, w_2) \quad (4.2.2)$$

for every $w_1, w_2 \in \mathcal{W}$.

By definition a Lagrangian subspace of a linear $2d$ dimensional symplectic space $\mathcal{W}$ is a maximal subspace on which ω vanishes (Lagrangian subspaces are d dimensional, see Problem 4.13).

¹²A symplectic manifold is a smooth manifold of even dimensions together with a symplectic form. By *symplectic form* we mean an antisymmetric differential two form ω which is closed and non degenerate. By “closed” we mean that $d\omega = 0$, by “non degenerate” that for all $v \in T_x X$ if $\omega(v, w) = 0$ for each $w \in T_x X$, then $v = 0$, see [5] for more details.

¹³Given a symplectic form ω on a manifold of dimension $2d$ the $2d$ form $\wedge^d \omega$ is a volume form: the symplectic volume.

¹⁴A map is symplectic if it conserves the symplectic two form ω , that is, for each $x \in X$ and vectors $v, w \in T_x X$, holds $\omega(D_x T v, D_x T w) = \omega(v, w)$.

Definition 4.2.2 *Given two transversal Lagrangian subspaces V_1 and V_2 we define the sector between V_1 and V_2 by*

$$\mathcal{C} = \mathcal{C}(V_1, V_2) = \{w \in \mathcal{W} \mid \omega(v_1, v_2) \geq 0 \text{ for } w = v_1 + v_2, v_i \in V_i, i = 1, 2\}$$

Equivalently, if we define the quadratic form associated with an ordered pair of transversal Lagrangian subspaces,

$$\mathcal{Q}(w) = \omega(v_1, v_2)$$

where $w = v_1 + v_2$, is the unique decomposition of w with the property $v_i \in V_i, i = 1, 2$, then we have

$$\mathcal{C} = \{w \in \mathcal{W} \mid \mathcal{Q}(w) \geq 0\}.$$

In the case of the standard symplectic space, $V_1 = \mathbb{R}^d \times \{0\}$ and $V_2 = \{0\} \times \mathbb{R}^d$ we get

$$\mathcal{Q}((\xi, \eta)) = \langle \xi, \eta \rangle$$

and

$$\mathcal{C}_+ = \{(\xi, \eta) \in \mathbb{R}^d \times \mathbb{R}^d \mid \langle \xi, \eta \rangle \geq 0\}.$$

We will refer to this $\mathcal{C}_+$ as the standard sector. Since any two pairs of transversal Lagrangian subspaces are symplectically equivalent (see Problem 4.15) we may consider only this case without any loss of generality.

It is natural to ask if a sector determines uniquely its sides. It is not a vacuous question since, for $d > 1$, there are many Lagrangian subspaces in the boundary of a sector. The answer is positive.

Proposition 4.2.3 *For two pairs of transversal Lagrangian subspaces V_1, V_2 and V'_1, V'_2 if*

$$\mathcal{C}(V_1, V_2) = \mathcal{C}(V'_1, V'_2)$$

then

$$V_1 = V'_1 \text{ and } V_2 = V'_2.$$

Moreover V_1 and V_2 are the only isolated Lagrangian subspaces contained in the boundary of the sector $\mathcal{C}(V_1, V_2)$.

Based on the notion of the sector between two transversal Lagrangian subspaces (or the quadratic form $\mathcal{Q}$) we define two monotonicity properties of a linear symplectic map. By $\text{int } \mathcal{C}$ we denote the interior of the sector, i.e.,

$$\text{int } \mathcal{C} = \{w \in \mathcal{W} \mid \mathcal{Q}(w) > 0\}.$$

Definition 4.2.4 *Given the sector $\mathcal{C}$ between two transversal Lagrangian subspaces we call a linear symplectic map L monotone if*

$$L\mathcal{C} \subset \mathcal{C}$$

and strictly monotone if

$$L\mathcal{C} \subset \text{int } \mathcal{C} \cup \{0\}.$$

A very useful characterization of monotonicity is given in the following

Proposition 4.2.5 *L is (strictly) monotone if and only if $\mathcal{Q}(Lw) \geq \mathcal{Q}(w)$ for every $w \in \mathcal{W}$ ($\mathcal{Q}(Lw) > \mathcal{Q}(w)$ for every $w \in \mathcal{W}$, $w \neq 0$). In particular, $\mathcal{Q}(Lw) = \mathcal{Q}(w)$, that is, L is a $\mathcal{Q}$ -isometry iff $L\mathcal{C} = \mathcal{C}$.*

The fact that monotonicity implies the increase of the quadratic form defining the cone is a manifestation of a very special geometric structure of a sector and does not hold for cones defined by general quadratic forms.

Proposition 4.2.6 *A monotone map L is strictly monotone if and only if*

$$LV_i \subset \text{int } \mathcal{C} \cup \{0\}, \quad i = 1, 2.$$

For the proofs of all the above facts see [94] and [64] or look at Problem 4.20, Problem 4.22 and Problem 4.24.

Remark 4.2.7 *Proposition 4.2.3 and 4.2.6 are trivial in the two dimensional case. As already notice, proposition 4.2.5 follows, in the two dimensional case, by (4.1.2).*

The relevance of the above discussion is the possibility to extend Theorem 4.1.1 to the present setting.

Theorem 4.2.8 (Wojtkowski [92]) *Let (X, μ, T) be a dynamical system where X is the finite union of Symplectic Manifolds, μ is the symplectic volume, T an invertible almost everywhere differentiable symplectic map of X and*

$$\int_X \log \|DT\| d\mu < \infty.$$

If there exists a measurable, a.e. non degenerate, eventually strictly invariant family of sectors then the Lyapunov exponents are different from zero almost surely.

PROOF. The proof follows the one of Theorem 4.1.1 where the algebraic considerations are replaced by Propositions 4.2.3, 4.2.5, 4.2.6, while the measure theoretical part is exactly the same. $\square$

4.2.1 Examples

4.2.1.a Linear symplectic maps

We will consider the following generalization of the Arnold cat. Let us consider the Dynamical Systems $(\mathbb{T}^{2d}, T, m)$, where m is the Lebesgue measure and $Tx = Lx \bmod 1$, with the following matrix L

$$L = \begin{pmatrix} \mathbb{1} & \mathbb{1} \\ M & \mathbb{1} + M \end{pmatrix}$$

where $M > 0$ and $M_{ij} \in \mathbb{Z}$ (see Problem 4.9 for a more concrete examples and Problem 4.21 to realize how general the example is). Then the system is symplectic and strictly monotone with respect to the standard sector, thus this Dynamical Systems is hyperbolic.

4.3 Metrics and cones

This section is devoted to a little digression on semi-metrics that can be associated to one sided convex cones. This is a rather vast field [90], here we will consider only few basic facts.

There exists a very geometric approach: consider the projectivization of the cone (that is the set of equivalence classes with respect to the equivalence relation $\sim$ defined by $v \sim w$ iff there exists $\lambda \in \mathbb{R}^+$ such that $v = \lambda w$) whereby obtaining a convex set in the projective space and then use the associated projective metric [28]. We will use a more direct, yet equivalent, approach (see Problem 4.28 and Problem 4.30 for further informations on the above point of view and the connection with the following). Hopefully, the reader will excuse the setting which is quite abstract in order to be applicable to some unexpected situations.

We start by illustrating some results in lattice theory originally due to Garrett Birkhoff. For more details see [12, 13], and [29], [68] for an overview of the field. Consider a topological vector space $\mathbb{V}$ with a partial ordering “ $\preceq$,” that is a vector lattice.¹⁵ We require the partial order to be *continuous*, i.e. given $\{v_n\} \in \mathbb{V}$ $\lim_{n \rightarrow \infty} v_n = v$, if $v_n \succeq w$ for each n , then $v \succeq w$. We call such vector lattices *integrally closed*.¹⁶

¹⁵We are assuming the partial order to be well behaved with respect to the algebraic structure: for each $v, w \in \mathbb{V}$ $v \succeq w \iff v - w \succeq 0$; for each $v \in \mathbb{V}$, $\lambda \in \mathbb{R}^+ \setminus \{0\}$ $v \succeq 0 \implies \lambda v \succeq 0$; for each $v \in \mathbb{V}$ $v \succeq 0$ and $v \preceq 0$ imply $v = 0$ (antisymmetry of the order relation).

¹⁶To be precise, in the literature “integrally closed” is used in a weaker sense. First, $\mathbb{V}$ does not need a topology. Second, it suffices that for $\{\alpha_n\} \in \mathbb{R}$, $\alpha_n \rightarrow \alpha$; $v, w \in \mathbb{V}$, if $\alpha_n v \succeq w$, then $\alpha v \succeq w$. Here we will ignore these and other subtleties: our task is limited to a brief account of the results relevant to the present context.

We define the closed convex cone¹⁷ $\mathcal{C} = \{v \in \mathbb{V} \mid v \neq 0, v \succeq 0\}$ (hereafter, the term “closed cone” $\mathcal{C}$ will mean that $\mathcal{C} \cup \{0\}$ is closed). Conversely, given a closed convex cone $\mathcal{C} \subset \mathbb{V}$, enjoying the property $\mathcal{C} \cap -\mathcal{C} = \emptyset$, we can define an order relation by (see Problem 4.27)

$$v \preceq w \iff w - v \in \mathcal{C} \cup \{0\}.$$

Henceforth, each time that we specify a convex cone we will assume the corresponding order relation and vice versa. The reader must therefore be advised that “ $\preceq$ ” will mean different things in different contexts.

It is then possible to define a projective metric Θ (Hilbert metric),¹⁸ in $\mathcal{C}$, by the construction:

$$\begin{aligned} \alpha(v, w) &= \sup\{\lambda \in \mathbb{R}^+ \mid \lambda v \preceq w\} \\ \beta(v, w) &= \inf\{\mu \in \mathbb{R}^+ \mid w \preceq \mu v\} \\ \Theta(v, w) &= \log \left[\frac{\beta(v, w)}{\alpha(v, w)} \right] \end{aligned} \tag{4.3.1}$$

where we take $\alpha = 0$ and $\beta = \infty$ if the corresponding sets are empty.

Lemma 4.3.1 *The function Θ is a semi-metric in $\mathcal{C}$.*

PROOF. Clearly $\Theta(v, w) = 0$ is equivalent to $v = \lambda w$ for some $\lambda \in \mathbb{R}^+$, also $\Theta(v, w) = \Theta(w, v)$ and the triangle inequality can be easily checked (see Problem 4.27). $\square$

The importance of the previous constructions is due, in our context, to the following theorem.

Theorem 4.3.2 ([12]) *Let $\mathbb{V}_1$, and $\mathbb{V}_2$ be two integrally closed vector lattices; $L : \mathbb{V}_1 \rightarrow \mathbb{V}_2$ a linear map such that $L(\mathcal{C}_1) \subset \mathcal{C}_2$, for two closed convex cones $\mathcal{C}_1 \subset \mathbb{V}_1$ and $\mathcal{C}_2 \subset \mathbb{V}_2$ with $\mathcal{C}_i \cap -\mathcal{C}_i = \emptyset$. Let Θ_i be the Hilbert metric corresponding to the cone $\mathcal{C}_i$. Setting $\Delta = \sup_{v, w \in L(\mathcal{C}_1)} \Theta_2(v, w)$ we have*

$$\Theta_2(Lv, Lw) \leq \tanh \left(\frac{\Delta}{4} \right) \Theta_1(v, w) \quad \forall v, w \in \mathcal{C}_1$$

¹⁷**Attention:** here, by “cone,” we mean any set such that, if v belongs to the set, then λv belongs to it as well, for each $\lambda > 0$. The reason for this change in the definition of cone is that two sided cones, viewed as sets, are never convex, while convexity plays a central rôle in the following. As we will see in Examples 4.3.1–Cones and Q -forms this change in definition does not limit the applicability of the present theory to the cones introduced in the previous section.

¹⁸In fact, we define a semi-metric, since $v \sim w \Rightarrow \Theta(v, w) = 0$. The metric that we describe corresponds to the conventional Hilbert metric on $\tilde{\mathcal{C}}$, the quotient of $\mathcal{C}$ with respect to the relation “ $\sim$ ”.

$(\tanh(\infty) \equiv 1)$.

PROOF. Let $v, w \in \mathcal{C}_1$. On the one hand if $\alpha := \alpha(v, w) = 0$ or $\beta := \beta(v, w) = \infty$, then the inequality is obviously satisfied. On the other hand, if $\alpha \neq 0$ and $\beta \neq \infty$, then

$$\Theta_1(v, w) = \ln \frac{\beta}{\alpha}$$

where $\alpha v \preceq w$ and $\beta v \succeq w$, since $\mathbb{V}_1$ is integrally closed. Notice that $\alpha \geq 0$, and $\beta \geq 0$ since $v \succeq 0$, $w \succeq 0$. If $\Delta = \infty$, then the result follows from $\alpha Lv \preceq Lw$ and $\beta Lv \succeq Lw$. If $\Delta < \infty$, then, by hypothesis,

$$\Theta_2(L(w - \alpha v), L(\beta v - w)) \leq \Delta$$

which means that there exist $\lambda, \mu \geq 0$ such that

$$\begin{aligned} \lambda L(w - \alpha v) &\preceq L(\beta v - w) \\ \mu L(w - \alpha v) &\succeq L(\beta v - w) \end{aligned}$$

with $\ln \frac{\mu}{\lambda} \leq \Delta$. The previous inequalities imply

$$\begin{aligned} \frac{\beta + \lambda\alpha}{1 + \lambda} Lv &\succeq Lw \\ \frac{\mu\alpha + \beta}{1 + \mu} Lv &\preceq Lw. \end{aligned}$$

Accordingly,

$$\begin{aligned} \Theta_2(Lv, Lw) &\leq \ln \frac{(\beta + \lambda\alpha)(1 + \mu)}{(1 + \lambda)(\mu\alpha + \beta)} = \ln \frac{e^{\Theta_1(v, w)} + \lambda}{e^{\Theta_1(v, w)} + \mu} - \ln \frac{1 + \lambda}{1 + \mu} \\ &= \int_0^{\Theta_1(v, w)} \frac{(\mu - \lambda)e^\xi}{(e^\xi + \lambda)(e^\xi + \mu)} d\xi \leq \Theta_1(v, w) \frac{1 - \frac{\lambda}{\mu}}{\left(1 + \sqrt{\frac{\lambda}{\mu}}\right)^2} \\ &\leq \tanh\left(\frac{\Delta}{4}\right) \Theta_1(v, w). \end{aligned}$$

□

Remark 4.3.3 In general, it suffices to know $L(\mathcal{C}_1) \subset \mathcal{C}_2$ in order to conclude $\Theta_2(Lv, Lw) \leq \Theta_1(v, w)$. However, a strict contraction depends on the diameter of the image being finite.¹⁹

¹⁹In the theory of Markov processes this corresponds to the so called *positivity improving* (see also Example 4.3.1.a).

In particular, if an operator maps a convex cone strictly inside itself (in the sense that the diameter of the image is finite), then it is a contraction in the Hilbert metric. This implies the existence of a “positive” eigenfunction (provided the cone is complete with respect to the Hilbert metric), and, with some additional work, the existence of a gap in the spectrum of L (see [12] for details or solve Problem 4.35).

Usually the space $\mathbb{V}$ comes endowed with its own metric, in such a case it is natural to wonder about the strength of the Hilbert metric compared to other metrics. While, in general, the answer depends on the cone, it is nevertheless possible to state an interesting result.

Definition 4.3.4 A function $\rho : \mathbb{V} \rightarrow \mathbb{R}^+$ is called homogeneous of degree one if for all $\lambda \in \mathbb{R}$ and $v \in \mathbb{V}$

$$\rho(\lambda v) = |\lambda| \rho(v).$$

Remark 4.3.5 Note that a norm or a linear functional are both homogeneous function of degree one.

Definition 4.3.6 A homogeneous function of degree one is called adapted to a cone $\mathcal{C}$ if, for each $v, w \in \mathbb{V}$,

$$-v \preceq w \preceq v \implies \rho(v) \geq \rho(w),$$

and $v \in \text{int } \mathcal{C}$ implies $\rho(v) > 0$.

Lemma 4.3.7 Let ρ_i be two homogeneous functions of degree one adapted to the cone $\mathcal{C} \subset \mathbb{V}$. Then, given $v, w \in \text{int } \mathcal{C} \subset \mathbb{V}$ for which $\rho_1(v) = \rho_1(w)$,

$$\rho_2(v - w) \leq \left(e^{\Theta(v, w)} - 1 \right) \min\{\rho_2(v), \rho_2(w)\}.$$

PROOF. We know that $\Theta(v, w) = \ln \frac{\beta}{\alpha}$, where $\alpha v \preceq w \preceq \beta v$. This implies that $-w \preceq 0 \preceq \alpha v \preceq w$, i.e. $\rho_1(w) \geq \alpha \rho_1(v)$, or $\alpha \leq 1$. In the same manner it follows that $\beta \geq 1$. Hence,

$$\begin{aligned} w - v &\preceq (\beta - 1)v \preceq (\beta - \alpha)v \\ w - v &\succeq (\alpha - 1)v \succeq -(\beta - \alpha)v \\ w - v &\preceq (1 - \beta^{-1})w \preceq (\alpha^{-1} - \beta^{-1})w \\ w - v &\succeq (1 - \alpha^{-1})w \succeq -(\alpha^{-1} - \beta^{-1})w \end{aligned}$$

which implies

$$\begin{aligned} \|w - v\| &\leq (\beta - \alpha)\|v\| \leq \frac{\beta - \alpha}{\alpha}\|v\| = \left(e^{\Theta(v, w)} - 1 \right) \|v\| \\ \|w - v\| &\leq (\alpha^{-1} - \beta^{-1})\|w\| \leq \left(\frac{\beta}{\alpha} - 1 \right) \beta^{-1} \|w\| \leq \left(e^{\Theta(v, w)} - 1 \right) \|w\|. \end{aligned}$$

□

Many normed vector lattices satisfy the hypothesis of Lemma 1.3 (e.g. Banach lattices²⁰).

4.3.1 Examples

4.3.1.a Perron-Frobenius Theorem

Consider a matrix $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of all strictly positive elements: $L_{ij} \geq \gamma > 0$. The Perron-Frobenius theorem states that there exists a unique eigenvector v^+ such that $v_i^+ > 0$, in addition the corresponding eigenvalue λ is simple, maximal and positive. There quite a few proofs of this theorem a possible one is based on Birkhoff theorem. Consider the cone $\mathcal{C}^+ = \{v \in \mathbb{R}^n \mid v_i \geq 0\}$, then obviously $L\mathcal{C}^+ \subset \mathcal{C}^+$. Moreover an explicit computation (see Problem 4.) shows that

$$\Theta(v, w) = \ln \sup_{ij} \frac{v_i w_j}{v_j w_i}. \quad (4.3.2)$$

Then, setting $M = \max_{ij} L_{ij}$, it follows that

$$\Theta(Lv, Lw) \leq 2 \ln \frac{M}{\gamma} := \Delta < \infty.$$

We have then a contraction in the Hilbert metric and the result follows from usual fixed points theorems. Note that, since $\Theta(v, \lambda v) = 0$, for all $\lambda \in \mathbb{R}^+$, the fixed point $v_+ \in \mathbb{R}^n$ is only projective, that is $Lv_+ = \lambda v_+$ for some $\lambda \in \mathbb{R}$; in other words, we have an eigenvalue.

Remark that L^* satisfies the same conditions as L , thus there exists $w^+ \in \mathcal{C}^+$, $\mu \in \mathbb{R}^+$, such that $L^*w^+ = \mu w^+$. Next, define $\rho_1(v) = |\langle w^+, v \rangle|$ and $\rho_2(v) = \|v\|$. It is easy to check that they are two homogeneous forms of degree one adapted to the cone.

In addition, if $\rho_1(v) = \rho_2(v)$, then $\rho_1(L^n v) = \rho_1(L^n w)$. Hence, by Lemma 4.3.7

$$\begin{aligned} \|L^n v - L^n w\| &\leq \left(e^{\Theta(L^n v, L^n w)} - 1 \right) \min\{\|L^n v\|, \|L^n w\|\} \\ &\leq K \Lambda^n \min\{\|L^n v\|, \|L^n w\|\}, \end{aligned} \quad (4.3.3)$$

for some constant K depending only on v, w . The estimate 4.3.3 means that all the vectors in the cone grow at the same rate. In fact, for all $v \in \text{int}\mathcal{C}$,

$$\|\lambda^{-n} L^n v - \lambda^{-n} L^n w\| \leq K \Lambda^n.$$

²⁰A Banach lattice $\mathbb{V}$ is a vector lattice equipped with a norm satisfying the property $\| |v| \| = \|v\|$ for each $f \in \mathbb{V}$, where $|v|$ is the least upper bound of v and $-v$. For this definition to make sense it is necessary to require that $\mathbb{V}$ is “directed,” i.e. any two elements have an upper bound.

Hence, $\lim_{n \rightarrow \infty} \lambda^{-n} L^n v = v_+$.

Finally, consider $\mathbb{V}_1 = \{v \in \mathbb{V} \mid \langle w^+, v \rangle = 0\}$. Clearly $L\mathbb{V}_1 \subset \mathbb{V}_1$ and $\mathbb{V}_1 \oplus \text{span}\{v_+\} = \mathbb{V}$. Let $w \in \mathbb{V}_1$, clearly there exists $\alpha \in \mathbb{R}^+$ such that $\alpha v_+ + w \in \mathcal{C}$,²¹ thus

$$\|L^n w\| \leq \|L^n(\alpha v_+ + w) - \alpha L^n v_+\| \leq L\Lambda^n \lambda^n.$$

This immediately implies that L restricted to the subspace $\mathbb{V}_1$ has spectral radius less than $\lambda\Lambda$. In other words, λ is the maximal eigenvalue, it is simple and any other eigenvalue must be smaller than $\lambda\Lambda$. We have thus obtained an estimate of the spectral gap between the first and the second eigenvalue.

4.3.1.b Cones and Q -forms

Here we would like to consider only half of the cone defined by the Q -form in order to apply the present theory. If $L_{ij} > 0$, then we can choose the first quadrant; on the other hand, if $L_{ij} < 0$, then L maps the first into the third quadrant. In both cases a monotone matrix L maps a one sided cone into a one sided cone. Here we will consider only the first case and leave the other—essentially identical—to the reader.²² Consequently, L is a monotone matrix with respect to the standard sector $\mathcal{C}_+$ and $L_{ij} > 0$, then $LC^+ \subset \mathcal{C}^+$ where, as in the previous example, $\mathcal{C}^+ = \{v \in \mathbb{R}^2 \mid v_i \geq 0\}$. Thus all the results of the previous example apply.

In particular, remembering (4.3.2), if $v = (1, \alpha)$, $w = (1, \beta) \in \mathcal{C}_+$, then

$$\Theta(v, w) = \left| \ln \frac{\alpha}{\beta} \right|. \quad (4.3.4)$$

Another interesting formula is (see Problem 4.34)

$$2 \sinh\left(\frac{1}{2}\Theta(v, w)\right) = \frac{|\omega(v, w)|}{\sqrt{Q(v)Q(w)}}. \quad (4.3.5)$$

This means that here exists a relation between the Hilbert metric and the Q -form.

To understand this relation better, let us compute

$$\text{diam}(LC^+) = \sup_{\alpha, \beta > 0} \left| \ln \frac{(a + \alpha b)(c + \beta d)}{(a + \beta b)(c + \alpha d)} \right|.$$

Since

$$\frac{a}{c} = \frac{ad}{cd} = \frac{1 + bc}{cd} = \frac{b}{d} + \frac{1}{cd} > \frac{b}{d}.$$

²¹this is a special case of the general fact that any vector can be written as the linear combination of two vectors belonging to the cone.

²²An easy way out is to consider L^2 instead of L .

Thus

$$\text{diam}(LC_+) = \left| \ln \frac{ad}{cb} \right| = \ln \frac{1+bc}{bc}, \quad (4.3.6)$$

which implies that, if L is strictly monotone, then $\text{diam}(LC^+) < \infty$. Accordingly, the rate of contraction of the Hilbert metric is given by

$$\Lambda = \frac{1 - \sqrt{\frac{bc}{1+bc}}}{1 + \sqrt{\frac{bc}{1+bc}}} = \frac{1}{(\sqrt{1+bc} + \sqrt{bc})^2} = \frac{1}{\sigma(L)^2}, \quad (4.3.7)$$

where the last equality follows by (4.1.2).

Remark 4.3.8 *It is not immediately clear how to extend the above considerations to the higher dimensional setting discussed in section 4.2. In fact, to do so it is necessary to introduce a different metric [63] of Caratheodory type [90]. We will not do it here but the reader should be aware that such a generalization is possible.*

4.3.1.c Expanding maps—uniqueness of the a.c. measure

A remarkable fact of Birkhoff theorem is that it applies to infinite dimensional vector spaces. In Example 1.4.1 we have studied the properties of $\mathcal{L}$. A computation similar to the one done there shows that, given a twice differentiable expanding map of the torus, the cone

$$\mathcal{C}_\alpha = \{h \in \mathcal{C}^{(0)}(\mathbb{T}) \mid h \geq 0; \frac{h(x)}{h(y)} \leq e^{\alpha d(x,y)}\} \quad (4.3.8)$$

is invariant. In fact, if $h \in \mathcal{C}_\alpha$

$$\begin{aligned} \mathcal{L}h(x) &= \sum_{z \in T^{-1}x} |D_z T|^{-1} h(z) \leq \sum_{w \in T^{-1}y} \frac{|D_w T|}{|D_z T|} |D_w T|^{-1} h(w) e^{\alpha d(z,w)} \\ &\leq \sum_{w \in T^{-1}y} |D_w T|^{-1} h(w) e^{(\lambda^{-1}\alpha + C)d(x,y)} = e^{(\lambda^{-1}\alpha + C)d(x,y)} \mathcal{L}h(y). \end{aligned}$$

By choosing α large enough, there exists $\sigma \in (\lambda^{-1}, 1)$ such that $\mathcal{L}\mathcal{C}_\alpha \subset \mathcal{C}_{\sigma\alpha}$.

A direct computation shows that the diameter is finite (see Problem 4.33). Accordingly, we have a contraction in the Hilbert metric. This implies that there exists only one invariant measure μ which is absolutely continuous with respect to Lebesgue ($d\mu = h_* dm$). Moreover, if $\rho_1(f) = \|\int f\|$ and $\rho_2(f) = \|f\|_\infty$, we have that Lemma 4.3.7 applies whereby showing that $\mathcal{L}h \rightarrow h_*$ in the sup norm for all $h \in \mathcal{C}_\alpha$, $\rho_1(h) = \rho_2(h_*)$. By arguments similar to the one employed

in 4.3.1.a it is possible to see that $\mathcal{L}$, viewed as an operator in $\mathcal{C}^{(1)}(\mathbb{T})$, has a maximal eigenvalue one while all the rest of the spectrum is separate by a gap (see Problem 4.35), clearly this implies not only the mixing but provides as well an estimate on the mixing rate for $\mathcal{C}^{(1)}(\mathbb{T})$ functions (see Problem 4.36).

4.4 Cones and invariant distributions

Here we use the machinery developed in the previous section to obtain a constructive proof of the existence of the unstable distribution in a special, but very interesting, case.

Lemma 4.4.1 *Given a smooth Symplectic Dynamical Systems with singularities (X, T, μ) , X a symplectic two dimensional manifold, μ the symplectic volume, if the systems is eventually strictly monotone, then $\{E^u(x)\}$ is almost everywhere well defined. Moreover, if the system is smooth and $\mathcal{C}(x)$ is continuous, then $\{E^u(x)\}$ is continuous (where it is defined). In addition, if the cone family is strictly monotone, then $\{E^u(x)\}$ is everywhere defined.*

The same holds for $\{E^s(x)\}$ and, in the last case, the two distributions are uniformly transversal.

PROOF. Let $\mathcal{C}_n(x) := D_{T^{-n}x} T^n \mathcal{C}(T^{-n}x)$ and $\Delta_n(x) := \text{diam}(\mathcal{C}_n(x))$, then Δ_n is decreasing, thus we can define

$$\Delta_\infty(x) := \lim_{n \rightarrow \infty} \Delta_n(x).$$

The key consequence of the results of section 4.3 (in particular Examples 4.3.1—Cones and Q -forms) is

$$\begin{aligned} \Delta_\infty(T^m x) &= \lim_{n \rightarrow \infty} \text{diam}(D_{T^{m-n}x} T^n \mathcal{C}(T^{-n+m}x)) \\ &= \lim_{n \rightarrow \infty} \text{diam}(D_x T^m D_{T^{-n}x} T^n \mathcal{C}(T^{-n}x)) \\ &\leq \frac{1}{\sigma(D_x T^m)^2} \Delta_\infty(x). \end{aligned}$$

Next, let $\Omega = \{x \in X \mid \Delta_\infty(x) = \infty\}$, we claim that $\mu(\Omega) = 0$. In fact, let $B_m = \{x \in X \mid \sigma(D_x T^m) \geq 2\}$, by eventual strict monotonicity of the cone field and Lemma 4.1.2 follows $\mu(\cup_{m \in \mathbb{N}} B_m) = \mu(X)$. In addition, $B_m \supset B_{m_0}$ for all $m > m_0$. Moreover, if $x \in B_m$, then $\Delta_\infty(T^m x) < \infty$ (see 4.3.7). Thus $T^{-n}\Omega \cap B_m = \emptyset$ for all $n \geq m$, and

$$\mu(\Omega) = \lim_{n \rightarrow \infty} \mu(T^{-n}\Omega) \leq \lim_{n \rightarrow \infty} \mu(X \setminus \cup_{m \leq n} B_m) = 0.$$

Finally, let $\Omega_L = \{x \in X \mid \frac{L}{2} \leq \Delta_\infty(x) \leq L\}$ and suppose $\mu(\Omega_L) > 0$. Then, there exists $n \in \mathbb{N}$ such that $\mu(\Omega_L \cap B_m) > 0$. Consequently, for almost all $x \in \Omega_L \cap B_m$ there exists a return time $\bar{n}m \in \mathbb{N}$ in the past (that is $T^{-\bar{n}m}x \in \Omega_L \cap B_m$). Accordingly,

$$\frac{L}{2} \leq \Delta_\infty(x) \leq \frac{\Delta_\infty(T^{-\bar{n}m}x)}{\sigma(D_x T^m)^2} \leq \frac{L}{4},$$

which is a contradiction unless $L = 0$. We have so proven that $\mu(\Omega_0) = \mu(X)$. In other words the cones $\mathcal{C}_\infty = \cap_{n \geq 0} \mathcal{C}_n(x)$ is almost everywhere degenerate since, having zero diameter, it consists of a single direction, such a direction is precisely the unstable direction.

To prove the continuity of the above distribution note that the cone family $\mathcal{C}_n(x)$ is continuous. Let x be such that $\Delta_\infty(x) = 0$, for each $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $\Delta_m(x) < \frac{\varepsilon}{2}$. Then one can chose δ such that the edges of $\mathcal{C}_m(y)$ vary, in the Hilbert metric, by an amount less than $\frac{\varepsilon}{2}$ if $d(x, y) < \delta$. This means that the distance between $E^u(x)$ and $E^u(y)$ is less than ε in the Hilbert metric of $\mathcal{C}(x)$, provided $d(x, y) < \delta$. In addition, all the unstable manifolds in such a neighborhood are uniformly in the interior of the cone $\mathcal{C}(x)$. The result follows then taking into account that the Hilbert metric is equivalent to the normal distance in each compact contained in the interior of the cone (see (4.3.4)).

The proof of the last fact is obvious: just a simplification of the above arguments. $\square$

With similar techniques it is also possible to construct the stable and unstable foliations, as we will see in chapter 7.

Let us conclude with an interesting simple fact.

Lemma 4.4.2 *A smooth two-dimensional Symplectic Dynamical System (X, T, μ) is Anosov iff it admits a strictly monotone continuous cone family.*

PROOF. By Lemma 4.4.1 it follows that the stable and unstable distribution are continuous. But then, by continuity, there exists $\alpha > 0$ and $\sigma > 1$ such that

$$\begin{aligned} \alpha \sqrt{Q(v)} &\leq \|v\| \leq \alpha^{-1} \sqrt{Q(v)} \quad \forall x \in X \text{ and } v \in E^u(x) \\ \sigma(D_x T) &\geq \sigma \quad \forall x \in X. \end{aligned}$$

Thus,

$$\|D_x T^n v\| \geq \alpha \sqrt{Q(D_x T^n v)} \geq \alpha \sigma^n \sqrt{Q(v)} \geq \alpha^2 \sigma^n \|v\|.$$

Analogously one can obtain the statement for the stable direction by using the cone family given by the complementary cones (see Problem 4.6).

The proof that an Anosov systems admit a continuous strictly invariant cone family is obvious and it is left to the reader.²³ $\square$

²³See Problem 7.5.

Theorem 4.4.3 *The set of Anosov two dimensional symplectic Dynamical Systems is open in the $\mathcal{C}^{(1)}$ topology.*

PROOF. Consider an Anosov map, let $\mathcal{C}$ be an associated continuous cone family. Clearly this will be a strictly invariant cone family also for any sufficiently close map. For more detail see Problem 4.7 and Problem 4.8. $\square$

In fact the above theorem can be easily generalized, see Problem 4.38.

Problems

- 4.1** Consider $\mathbb{R}^2$ endowed with the scalar product $\langle v, w \rangle_G := \langle v, Gw \rangle$, where $\langle \cdot, \cdot \rangle$ is the standard scalar product and $G > 0$. Show that there exists a change of coordinates $M : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that, in the new coordinates $\langle \cdot, \cdot \rangle_G$ becomes the standard scalar product.
- 4.2** Consider the cone $\mathcal{C}$ defined by the two transversal vectors $v_1, v_2 \in \mathbb{R}^2$. This means that $v \in \mathbb{R}^2$ belongs to the cone iff $v = \alpha v_1 + \beta v_2$ with $\alpha\beta \geq 0$. Show that there is a linear change of coordinates $M : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $M\mathcal{C} = \mathcal{C}_+$ and $\det M = 1$.
- 4.3** Show that the hypothesis of Theorem 4.1.1 can be relaxed, in particular it holds for smooth systems with singularities (see section 3.6). (Hint: Just follow the proof step by step and notice that nothing substantial need to be changed.)
- 4.4** Construct a strictly invariant cone family for the irrational translation on $\mathbb{T}^2$ (see Examples 1.1.1) and show that it is not measurable. (Hint: For each trajectory choose a point x . At such a point choose the standard cone $\mathcal{C}_+$, let $\mathcal{C}_n^- = \{(v_1, v_2) \in \mathbb{R}^2 \mid 1 + \frac{1}{n} \leq \frac{v_2}{v_1} \leq 2 + \frac{1}{n}\}$ and $\mathcal{C}_n^+ = \{(v_1, v_2) \in \mathbb{R}^2 \mid -2 - \frac{1}{n} \leq \frac{v_2}{v_1} \leq -1 - \frac{1}{n}\}^c$. Then set $\mathcal{C}(T^n x) = \mathcal{C}_n^+$ and $\mathcal{C}(T^{-n} x) = \mathcal{C}_n^-$. Such a cone family is strictly monotone by construction (since $D_x T = 1$), yet the system has obviously zero Lyapunov exponents. Since all the other hypothesis of Theorem 4.1.1 are satisfied, it follows that the above cone family cannot be measurable.)
- 4.5** Show that for two dimensional symplectic maps the sum of the Lyapunov exponent is zero (*pairing of the Lyapunov exponents*). (Hint: If $\omega(v, w) = 1$ then it holds that $1 = \omega(DT^n v, DT^n w) \sim \|DT^n v\| \|DT^n w\|$.)
- 4.6** Check that $\inf_{v \in \mathcal{C}_+} \sqrt{\frac{Q(Lv)}{Q(v)}} = \inf_{v \in \mathcal{C}_-} \sqrt{\frac{Q(L^{-1}v)}{Q(v)}}$, remember that $\mathcal{C}_- = \overline{(\mathcal{C}_+)^c}$. (Hint: see [63])

- 4.7** Show that, in a two dimensional area preserving systems, if the LE are different from zero then there exists and eventually strictly invariant cone family. (Hint: By Oseledec and Problem ?? there exists the unstable distributions, then construct the cones around it.)
- 4.8** Show that, Anosov two dimensional area preserving systems form an open set in the $\mathcal{C}^{(1)}$ topology in the set of area preserving diffeo.
- 4.9** Prove that if M is the two by two matrix

$$M = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

with $a, b, c \in \mathbb{Z}$, then $M > 0$ iff $a, c > 0$ and $c > \frac{b^2}{a}$.

- 4.10** Show that a $2d \times 2d$ matrix L of the form

$$L = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where A, B, C, D are $d \times d$ blocs, is symplectic, iff $C^*A = A^*C$, $D^*B = B^*D$ and $A^*D - C^*B = \mathbb{1}$. (Hint: Note that, by introducing the matrix

$$J = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}$$

the standard symplectic form 4.2.1 can be written in terms of the usual scalar product as

$$\omega((\xi^1, \xi^2), (\eta^1, \eta^2)) = \langle (\xi^1, \xi^2), J(\eta^1, \eta^2) \rangle.$$

From this point of view the definition of symplectic matrix 4.2.2 can be written as

$$L^*JL = J.$$

A trivial algebraic computation yields now the result.)

- 4.11** Prove that if L is symplectic then $\det L = 1$. (Hint: The determinant of a matrix is nothing else than the volume of the parallelepiped of sides $(Le_1, \dots, Le_{2d})$ (where $e_1, \dots, e_{2d}$ is the standard orthonormal basis of $\mathbb{R}^{2d}$). On the other hand the volume form can be written as $\wedge^d \omega$ (since that is a $2d$ form with the right normalization and the space of $2d$ forms is one dimensional). Thus $\det L = \wedge^d \omega(Le_1, \dots, Le_{2d}) = \wedge^d \omega(e_1, \dots, e_{2d}) = 1$ where we have used the fact that $\omega(Lv, Lu) = \omega(v, u)$. The reader that wants to appreciate the power of the above geometrical interpretation of the determinant and of the external forms can try to prove the statement by purely algebraic means.)

- 4.12** Given a subspace $E \in \mathbb{R}^{2d}$ we define the skew orthogonal subspace $E^\perp = \{v \in \mathbb{R}^{2d} \mid \omega(w, v) = 0 \ \forall w \in E\}$. Show that $E^\perp$ is a vector space and that $\dim(E) + \dim(E^\perp) = 2d$. (Hint: Given a basis $\{w_i\}$ of E , $\ell = \dim(E)$, consider the map $F : \mathbb{R}^{2d} \rightarrow \mathbb{R}^\ell$ defined by

$$F(v) := (\omega(w_1, v), \dots, \omega(w_\ell, v)).$$

Then $\text{Ker}(F) = E^\perp$. Hence $\dim E^\perp = 2d - \dim R(F)$. Clearly $\dim R(F) \leq \ell$, in addition if $R(F) < \ell$ then there must exist a vector $(\alpha_1, \dots, \alpha_\ell) \in \mathbb{R}^\ell$ such that for each $v \in \mathbb{R}^{2d}$

$$0 = \sum_i \alpha_i \omega(w_i, v) = \omega\left(\sum_i \alpha_i w_i, v\right)$$

which, by the non degeneracy of ω , would imply $\sum_i \alpha_i w_i = 0$, a contradiction.)

- 4.13** Show that Lagrangian subspaces are d dimensional. (Hint: Let E be a Lagrangian subspace, then $E^\perp \supset E$. Suppose there exists $v \in E^\perp \setminus E$, then for each $w_i = \alpha_i v + \bar{v}_i$, $i \in \{1, 2\}$, $\bar{v}_i \in E$

$$\omega(w_1, w_2) = \omega(\alpha_1 v + \bar{v}_1, \alpha_2 v + \bar{v}_2) = 0.$$

Thus E is not maximal. Accordingly $E^\perp = E$ and the result follows by Problem 4.12.)

- 4.14** Show that all the Lagrangian subspaces transversal to $V = \{(0, \eta) \in \mathbb{R}^{2d} \mid \eta \in \mathbb{R}^d\}$ can be represented as $\{(\xi, U\xi \in \mathbb{R}^{2d} \mid \xi \in \mathbb{R}^d\}$ for some symmetric matrix U . (Hint: Let $V_U := \{(\xi, U\xi \in \mathbb{R}^{2d} \mid \xi \in \mathbb{R}^d\}$, then $\omega((\xi, U\xi), (\zeta, U\zeta)) = 0$, thus V_U is Lagrangian. On the other hand, if $\tilde{V}$ is Lagrangian, then it is a d dimensional space (Problem 4.13). Let $\{(\xi_i, \eta_i)\}_{i=1}^d$ be a base for $\tilde{V}$, then $\xi_i \neq 0$ by the transversality assumption and we can define the matrix U via $U\xi := \eta_i$. It is immediate that $\tilde{V}$ Lagrangian implies $U = U^*$.)

- 4.15** Show that any two transversal Lagrangian subspaces can be mapped into $\{(u, 0)\}$ and $\{(0, u)\}$, respectively, by a symplectic transformation.

- 4.16** Show that all symplectic Q -isometries L (that is $Q(Lv) = Q(v)$) have the form

$$L = \begin{pmatrix} A & 0 \\ 0 & A^{*-1} \end{pmatrix}.$$

(Hint: Start by considering the vector $(0, u)$, $U \in \mathbb{R}^d$, clearly $Q((0, u)) = 0$ thus $Q(L(0, u)) = 0$ if L is a Q -isometry. But if

$$L = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

it follows $\langle Bu, Du \rangle = 0$ for each $u \in \mathbb{R}^d$, that is $B^*D = 0$. The same argument applied to the vector $(u, 0)$ yields $A^*C = 0$. Accordingly, by symplecticity (see Problem 4.10),

$$\begin{aligned} Q(L(v, u)) &= \langle Au + Bv, Cu + Dv \rangle = \langle u, (A^*D + C^*B)v \rangle \\ &= \langle u, (\mathbb{1} + 2C^*B)v \rangle \end{aligned}$$

thus $Q(L(v, u)) = Q(v, u)$ iff $C^*B = 0$ which implies $A^*D = \mathbb{1}$.

4.17 Show that if the matrix

$$L = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

is symplectic then

$$L^{-1} = \begin{pmatrix} D^* & -B^* \\ -C^* & A^* \end{pmatrix}$$

(Hint: multiply and use Problem 4.10.)

4.18 Show that the symplectic matrices form a multiplicative group. (Hint: Use the definition and the above problems.)

4.19 A symplectic map L is a Q -isometry iff $LC = C$. (Hint: One direction is trivial. On the other hand, if $LC = C$ it follows that L maps the boundary, of C , to the boundary. Accordingly, if $\langle v, u \rangle = 0$ it must be

$$0 = \langle Av + bu, Cv + Du \rangle. \quad (4.4.1)$$

Choosing in 4.4.1 $u = 0$ yields $A^*C = 0$, choosing $v = 0$ shows that it must be $B^*D = 0$. Thus 4.4.1 yields

$$0 = \langle u, (A^*D + C^*B)v \rangle = 2\langle u, C^*Bv \rangle.$$

The above equality shows that C^*Bv is parallel to v for each $v \in R^d$, that is $C^*B = \alpha\mathbb{1}$ for some $\alpha \in \mathbb{R}$. If $\alpha = 0$, then $A^*D = \mathbb{1}$ and thus $C = 0$ which is the wanted result. If $\alpha \neq 0$, then B is invertible and $C = \alpha B^{*1}$. But this implies $A = 0$ and hence $-\mathbb{1} = C^*B = \alpha\mathbb{1}$, that is $\alpha = -1$. Accordingly the matrix would have the form

$$L = \begin{pmatrix} 0 & B \\ -B^{*-1} & 0 \end{pmatrix}$$

which sends C in its complement, contrary to our requirement.)

4.20 Prove Proposition 4.2.3.

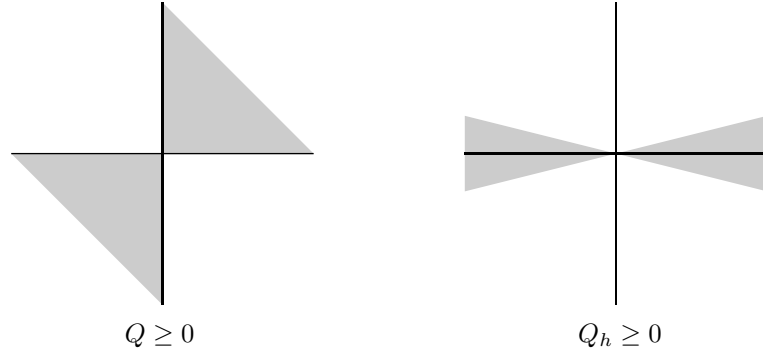


Figure 4.1: Cones

- 4.21** Show that a strictly monotone symplectic matrix can be put into the form

$$\begin{pmatrix} \mathbf{1} & \mathbf{1} \\ M & \mathbf{1} + M \end{pmatrix}$$

by multiplying it by Q -isometries on the left and on the right.

- 4.22** Prove Proposition 4.2.5.

- 4.23** Show that $V_U := \{(\xi, U\xi \in \mathbb{R}^{2d} \mid \xi \in \mathbb{R}^d\}$, $U = U^*$, belongs to the standard cone iff $U \geq 0$.

- 4.24** Prove Proposition 4.2.6.

- 4.25** Find a symplectic change of coordinates that transforms the standard form Q into the form Q_h defined by:

$$Q_h((x, y)) = \frac{1}{2}(h^2\langle x, x \rangle - \langle y, y \rangle),$$

and draw the associate cone. (Hint: Consider

$$\begin{aligned} x &= hx' - y' \\ y &= hx' + y'. \end{aligned}$$

See the figure for the shape of the cones.)

- 4.26** Do the same as in Problem Problem 4.25 but with the cone $\{(v_1, v_2) \in \mathbb{R}^2 \mid \alpha \leq \frac{|v_1|}{|v_2|} \leq \alpha^{-1}\}$. (**make picture**

- 4.27** Given a cone $\mathcal{C}$ in a vector space $\mathbb{V}$ define the relation $v \succeq w$ if $v - w \in \mathcal{C}$. Prove that $v \succeq w$ and $w \succeq z$ implies $v \succeq z$ if and only if $\mathcal{C}$ is convex.

Show that $v \succeq w$ and $w \succeq v$ implies $v = w$ if and only if $\mathcal{C} \cup (-\mathcal{C}) = \emptyset$. If $\mathcal{C}$ has the two above properties, then $\succeq$ is an order relation and $\mathbb{V}$ equipped with it is a Lattice

- 4.28** Let $C \in \mathbb{R}^n$ be a strictly convex compact set. For each two point $x, y \in C$ consider the line $\ell = \{\lambda x + (1 - \lambda)y \mid \lambda \in \mathbb{R}\}$ passing through x and y . Let $\{u, v\} = C \cap \ell$ and define

$$\Theta(x, y) = \left| \ln \frac{\|x - u\| \|y - v\|}{\|x - v\| \|y - u\|} \right|$$

(the logarithm of the cross ratio). Show that Θ is a metric in C (the Hilbert metric). Show that the distance of $x \in \text{int } C$ from ∂C is infinite. (Hint: The only non trivial task is to check the triangle inequality. Consider three points $x, y, z \in C$. If the points are collinear then the proof is easy. If they are not the consider the plane defined by them, we have now a two dimensional problem, thus it suffices to prove the result in $\mathbb{R}^2$. Consider the Figure 4.2 and remember that the cross ratio

$$R(x, y, u, v) = \frac{\|x - u\| \|y - v\|}{\|x - v\| \|y - u\|}$$

is a projective invariant. Then

$$\begin{aligned} R(x, z, u, v) &= R(x, w, x', y') \geq R(x, w, \alpha, \beta) \\ R(y, z, a, b) &= R(w, y, x', y') \geq R(w, y, \alpha, \beta) \end{aligned}$$

and the result follows since $R(x, w, \alpha, \beta) R(w, y, \alpha, \beta) = R(x, y, \alpha, \beta)$.)

- 4.29** Prove the same as the previous Problem for convex sets in a projective space.
- 4.30** Check that the metric defined in Problem 4.28 and Problem 4.29 applied to the projectivization of convex cones yields to the same metric defined in (4.3.1). (Hint: Let $\mathcal{C} \subset \mathbb{V}$ be a convex cone. The projectivization consists in considering the space $\mathbb{P}$ of the equivalence classes $[v]$ with respect to the equivalence relation $v \sim w$ iff $v = \lambda w$ for some $\lambda \in \mathbb{R}^+$. Let $\tilde{\mathcal{C}} = \{[v] \in \mathbb{P} \mid v \in \mathcal{C}\}$. Clearly $\tilde{\mathcal{C}}$ is convex in the projective space $\mathbb{P}$.²⁴ So, if $[x], [y] \in \tilde{\mathcal{C}}$, the Hilbert metric is defined by the points $[u], [v] \in \partial \tilde{\mathcal{C}}$ on the line determined by $[x], [y]$. By normalizing properly one obtains $[u] = [-\alpha x + y]$ and $[v] = [\beta x - y]$ from which the result follows.)
- 4.31** Compute the Hilbert metric for a disc and discuss its relation to the Poincaré metric (Hint: same?)

²⁴The lines in $\mathbb{P}$ are given by $[\alpha v + \beta w]$ where $\alpha, \beta \in \mathbb{R}$ where $[v] \neq [w]$.

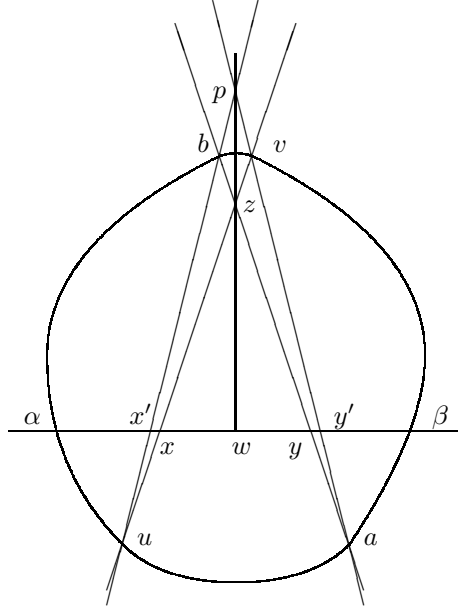


Figure 4.2: Hilbert metric

4.32 Kobayashi and Caratheodory metrics....

4.33 Show that the Hilbert metric for the cone $\mathcal{C}_\alpha$ defined in (4.3.8) is given by ???

4.34 Prove formula (4.3.5).

4.35 Show that the Perron-Frobenius operator associated to a smooth expanding map of the circle has a spectral gap as an operator on $Lip(\mathbb{T}^2)$. (Hint: Check that there exists $b \in \mathbb{R}^+$ such that the norm

$$\|h\| := \|h\|_\infty + b\|h\|_{Lip}$$

is adapted to the cone. Define $\mathbb{V} = \{h \in Lip(\mathbb{T}^2) \mid \int h = 0\}$, notice that $\mathcal{L}\mathbb{V} = \mathbb{V}$. Then, for each $h \in \mathbb{V}$ there exists $\rho \in \mathbb{R}^+$ such that $h + \rho h_* \in \mathcal{C}_\alpha$, so

$$\|\mathcal{L}^n h\| = \|\mathcal{L}^n(h + \rho h_*) - \rho h_*\| \leq K\Lambda^n \rho.$$

Thus the spectral radius of $\mathcal{L}|_{\mathbb{V}}$ is less than Λ .)

4.36 Estimate the rate of mixing for Lipschitz functions for a smooth expanding map of the circle (Hint: use the spectral gap of the previous Problem.)

4.37 Prove that any continuous fraction of the form

$$\frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots}}}$$

$a_i > 0$ is convergent provided the series $\sum_{n=1}^{\infty} a_n$ is divergent. (Hint: Let

$$\prod_{i=1}^n \begin{pmatrix} 1 & a_{2(n-i)} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ a_{2(n-1)+1} & 1 \end{pmatrix} \begin{pmatrix} 1 \\ u \end{pmatrix} = \begin{pmatrix} \beta_n \\ \alpha_n \end{pmatrix}$$

and verify, by induction, that $\frac{\alpha_n}{\beta_n}$ is exactly the $2n$ truncation of the continuous fraction. Thus the continuous fraction is a projective coordinate for the vector (α_n, β_n) . Consider the cone $\mathcal{C}_+ = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0; y \geq 0\}$. Then, for each $a, b \in \mathbb{R}^+$, holds

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} \mathcal{C}_+ \subset \mathcal{C}_+.$$

The result follows by computing the Hilbert metric contraction.

For a different approach see [91][Th14.1].)

4.38 Show that, Anosov systems form an open set in the $\mathcal{C}^{(1)}$ topology.

Notes

When cones first appeared?

The point of W. th. is that no estimate on the cone contraction is needed. To appreciate the advantage one can try to prove the existence of the LE via direct estimates for the Levovich map in example 4.1.1.c

Generalization of metrics on cones, see [90]